



Public Review for

LEOScope: Building a Global Testbed for Low-Earth Orbit Satellite Networks

Saeed Fadaei, Shubham Tiwari, Aryan Taneja, Saksham Bhushan,
Mohamed Kassem, Aravindh Raman, Debopam Bhattacharjee, Lili
Qiu, Alan Woodward, Nishanth Sastry

This paper presents LEOScope, a global testbed to quantify the performance and bottleneck of the operational Low Earth Orbit (LEO) satellite network. LEOScope incorporates several features to facilitate LEO network measurements, such as docket-based deployment, event-driven trigger mode, and various safeguards against misuse and disruptions of network probes. The paper showcases LEOScope's effectiveness in measuring Starlink's performance.

While the reviewers were overall very positive about this contribution, several areas required improvements for the final version of the paper: (1) novelty - clarifying new technical contributions from RIPE, Hypatia, Edgenet or others, (2) usage considerations - discussing pricing, security, fair use and effect on clients' network, (3) extending on new technical challenges addressed by LeoScope, and (4) discussing expected costs and requirements from new volunteers, and how this work hopes to lower barrier to entry.

Overall, the reviewers believe that LEOScope can boost the community efforts for large-scale LEO network measurements, making it a worthy contribution to the research community.

Public review written by
Noa Zilberman
University of Oxford, UK

LEOScope: Building a Global Testbed for Low-Earth Orbit Satellite Networks

Saeed Fadaei
University of Surrey
s.fadaei@surrey.ac.uk

Shubham Tiwari*
University of Washington
tshubham@cs.washington.edu

Aryan Taneja*
UIUC
aryant2@illinois.edu

Saksham Bhushan
Microsoft Research
saksham291@gmail.com

Mohamed Kassem
University of Surrey
m.m.m.kassem5@gmail.com

Aravindh Raman[†]
Cisco ThousandEyes
araram@cisco.com

Debopam
Bhattacharjee
Microsoft Research
debopamb@microsoft.com

Lili Qiu
Microsoft Research
liliqiu@microsoft.com

Alan Woodward
University of Surrey
alan.woodward@surrey.ac.uk

Nishanth Sastry
University of Surrey
n.sastry@surrey.ac.uk

ABSTRACT

Low-Earth Orbit (LEO) satellite mega-constellations which offer broad coverage and low latencies offer a revolutionary new connectivity option. However, they also have complex orbital dynamics leading to continuous latency changes and frequent satellite hand-offs. We are building a globally spanning testbed, LEOScope¹, to quantify the performance opportunities and bottlenecks of such networks. This paper discusses the unique features we incorporated into LEOScope to adapt it to the dynamics of LEO networks and to enable experimentation on volunteer nodes. To demonstrate the broad utility of the testbed, we report Starlink performance across countries using LEOScope. Our primary aim is to describe the testbed and announce its availability to the SIGCOMM community for non-commercial research activities.

CCS CONCEPTS

• **Networks** → **Network measurement; Network experimentation; Network performance analysis; Programming interfaces;**

KEYWORDS

satellite networks, LEO constellations, network testbed, performance measurement, orbital dynamics

1 INTRODUCTION

Recent years have marked an increased adoption of satellite Internet driven by new developments in Low Earth Orbit (LEO) satellite networks. Major players like SpaceX [54], Amazon [13], and Eutelsat OneWeb [47] have heavily invested and engineered satellite *mega-constellations* to deliver high-speed Internet access to anywhere in the world. This has led to numerous measurement studies [16–18, 22, 24, 26–28, 30, 31, 33, 34, 38, 40, 42, 45, 52, 58, 59], but each has utilised various point solutions, such as deploying isolated test

nodes for active measurements in specific regions [40, 42, 52, 61], gathering data from public measurement platforms intended for different purposes [45, 50], or simulating satellite networks using tools like Hypatia [33], Starrynet [35] or xeoverse [32].

While these point solutions offer valuable insights, their take-aways are affected by certain factors. Firstly, isolated measurements present logistical challenges in replicating results, limiting their generalizability to specific areas or user bases. Secondly, data-based analyses are less controlled and essential parameters are anonymized for privacy reasons, which can affect the accuracy and reliability of the conclusions drawn. Thirdly, simulations are effective for modelling scenarios but may not fully capture the complexities and nuances of actual satellite network performance.

To address the above concerns and as a follow-up to the SatNetLab proposal [53], we introduce LEOScope – an extensible LEO satellite network testbed to fully understand network performance across geographies and innovate at scale. We envision it to become the go-to platform for running customized network and application experiments over LEO networks. LEOScope complements existing terrestrial Internet testbeds by offering unique features that are specifically useful to understand the LEO subtleties.

To exercise the global Starlink network, we need a wide geographic footprint, and therefore need to be flexible enough to accommodate a variety of nodes that may wish to join. This currently consists of both nodes in volunteers’ homes as well as in universities in 7 nations across Europe, Africa and North America (See Appendix Fig. 4 map). Volunteer nodes are important to measure performance in remote locations where there may not be universities which can host a node.

We also made an explicit design choice to enable a wide variety of user-defined experiments, unlike several other widely used testbeds that run mainly on the terrestrial Internet and only allow tightly controlled types of experiments. To enable this flexibility, LEOScope runs experiments as docker containers; thus almost any functionality that can be wrapped up as a docker container can be exercised on the testbed. For example, an experiment can specify a regular ping every hour on the hour from a node in Africa to a

*Work done while the author was at Microsoft Research

[†] Author also affiliated with the University of Surrey. Work done while the author was at Telefonica Research.

¹<https://leoscope.surrey.ac.uk>

node in North America. Another experiment could test connectivity (latency; download speed etc) from selected nodes to the nearest instance of different CDN providers [19]. We believe this ability to run arbitrary experiments is essential at this initial stage of LEO megaconstellation measurements, as we are only starting to understand their performance since the past two years [31, 42], and we may not have captured all the important and interesting types of measurements needed to create a useful closed catalog of canned experiments.

In addition, to fully capture the *dynamics* of LEO networks, experiments may need to run *in response to external triggers*, both networking-related (latency spike, throughput dip, satellite hand-offs, etc.) as well as non-networking events such as rain, snow, solar storms etc. To support this, we developed a *trigger mode* that schedules experiments based on user-specified conditions. LEOScope also includes capabilities to schedule experiments at regular intervals (similar to cronjobs) as well as flexibly at the first available time slot.

However, with all this flexibility of experimentation comes the possibility of intentional misuse or unintentional disruptions to the nodes hosting the experiment, which we cannot afford as volunteers may use Starlink as their primary or only source of connectivity. To engender a culture of responsible research, we incorporate both technical and non-technical safeguards. As the primary technical safeguard, we built a *scavenger mode*, which allows users (volunteers) to run their own networked applications without being affected by LEOScope’s scheduled experiments. Essentially, LEOScope continuously senses whether volunteers are using the network for their own (non-experiment) needs and reschedules experiments to a later time to avoid disrupting them.

We complement this with non-technical safeguards that ensure that the experimenters do not endanger the nodes (e.g., by accessing illegal content): Experiments are required to upload a proof of Institutional Review Board (IRB) approval. To make this process lightweight, we have designed a check list of questions that, if answered in the negative, provides a reasonable assurance that the experiment is “harmless” and can therefore be deployed without a full-fledged IRB process. Experimenters also need to sign up to an End User License Agreement (EULA) which has been carefully designed to correctly apportion responsibility between experimenters and the platform hosts (us).

To illustrate the possibilities of LEOScope, we demonstrate a few use cases. While we leave it to the experimenters’ imaginations to come up with an exhaustive list of interesting measurements possible at scale using the features above, we show how we used LEOScope to gather Starlink network performance (latency, throughput) insights across geographies. While LEOScope already has a few academic and industry research labs contributing resources and running experiments, with this work we aim to generate community excitement toward making LEOScope scale. *In essence, this paper is intended to serve as a call to the community of interested researchers to contribute nodes and use the testbed, as the number of LEO measurement studies burgeons [30, 31, 39, 40, 42, 45, 49, 50, 57].* The Appendices provide details for both researchers and volunteers on how to use LEOScope and contribute nodes to it.

2 RELATED WORK

Most existing testbeds are focused on terrestrial networks [14] and are not able to detect and react to LEO dynamics. While LEO broadband is still in its infancy, it is critical to quantify and understand its nuances well so that novel solutions could be developed across the network stack to unleash its full potential.

PlanetLab [21] served as a platform for development and testing of new network protocols and services on a global scale for the past two decades. Our objective is similar in spirit: to establish a “PlanetLab for LEO broadband” with a widespread presence across multiple regions. Similar to other prominent measurement platforms [20], we share with PlanetLab the ambition to host arbitrary experiments (as docker containers). However, the unique aspects of LEO gives rise to significant design differences (§3).

The RIPE Atlas [10] project aims at measuring global Internet connectivity and reachability through clients (probes) and measurement servers (anchors). Indeed, many probes exist behind Starlink terminals and have been used by researchers for Starlink measurements ([45, 50]). The probes, deployed by volunteers, can mostly run a restricted but standard set of network experiments like traceroute, ping, DNS, NTP, etc. LEOScope is flexible in onboarding new/innovative experiments subject to agreements.

M-Lab [23] runs a similarly fixed set of carefully vetted experiments, mainly implemented on the server side (for example, the network diagnostic tool or NDT [44]). While the M-lab approach is easily deployable since users bring their own devices, the experiments are only conducted when a user accesses the tool and opts to initiate a test. LEOScope’s trigger mode offers an innovative way to specify a rich set of triggers that could drive network performance.

LEOScope is similar to efforts such as BISmark [56] and SamKnows [11] which conduct broadband measurements by sending specially instrumented WiFi routers to homes and volunteers. However, unlike them, LEOScope is actively designed to incorporate arbitrary experiments. Testbeds specific to wireless and cellular settings [1, 5, 15, 25, 46, 48] have been running successfully for a while, but none of them are tuned to LEO’s dynamics. LEOScope, on the other hand, is the first measurement platform specifically designed for LEO satellite network research, distinguishing itself from general-purpose platforms like RIPE Atlas [2] and M-Lab [23] through integration of network performance measurements with satellite positioning data. Unlike PlanetLab [8], EdgeNet [20], and Monroe [12], LEOScope offers satellite-specific telemetry, event-based triggering for precise measurement of phenomena like handovers, resource-aware operation to ensure non-interference with volunteers’ networks, and hardware flexibility, addressing a critical gap in existing research infrastructure with minimal footprint while being able to take control of the intended LEO measurements.

Table 1 compares LEOScope with other prominent testbeds (more details in Appendix D for interested readers).

3 DESIGN CONSIDERATIONS

LEOScope is intended to be used by networking researchers for a diverse set of use cases, ranging from measuring how LEO constellations support common applications such as video streaming or teleconferencing, to designing new variants of existing end-to-end

Table 1: Comparison of Well-Known Testbeds used for network measurements

	Years of operation	Hardware Independent	Inbuilt telemetry	User-defined experiments	Internal and External Triggers	Scavenger Mode	Data access	Purpose
RIPE Atlas [10]	1992-	✗	✓	✗	✗	✗	Public	Broadband Measurements
M-LAB [23]	2010-	✓	✗	✗	✗	✗	Public	End-user Performance
PlanetLab [8]	2002-20	✓	✗	✓	✗	✗	Public	Network Service Deployments
Monroe [12]	2016-	✗	✓	✓	✗	✗	Public	Cellular Network
Samknows [11]	2008-	✗	✓	✗	✗	Partial	Restricted	Broadband Measurements
EdgeNet [20]	2021-	✓	✗	✓	✗	Partial	Restricted	Distributed Edge/ Cloud Development
LEOScope	2022-	✓	✓	✓	✓	✓	Public	Satellite Network Measurements

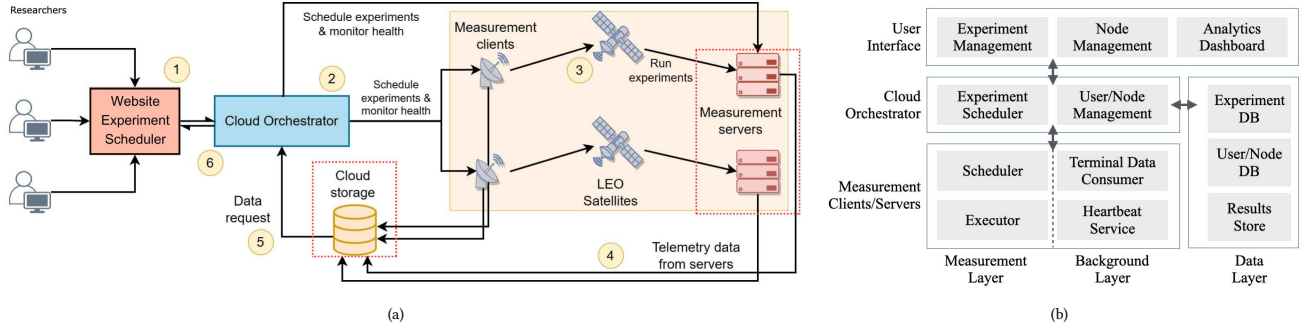


Figure 1: (a) highlights the building blocks of LEOScope and how they interact together. (b) shows the details and sub-modules of building blocks.

protocols (or even entirely new network stacks) that are adapted to the needs and functionalities of LEO networks.

To support this, LEOScope's design is driven by two key needs of LEO network measurements: diversity in geographic coverage and the dynamicity of LEO megaconstellations. It is now widely recognised [60] that LEO performance may vary over different latitudes – Starlink, for example, has less satellite density in higher latitudes. To accommodate such differences, *any new network protocol or measurement of existing applications needs to be validated over different geographies*. Even in a single location, the performance achieved highly depends on the numbers of satellites overhead at a given time point, which of those satellites the terminal is connected to, the Signal-to-Noise (SNR) achieved over that link, distance to the closest terrestrial gateway, and other factors. Exogenous factors such as adverse weather (rain, snow or cloud cover) can also affect the channel conditions [31, 36, 37]. Thus, researchers need to be able to explore resilience to or exploit the effects of such dynamicity in LEO networks.

To support geographic diversity, and especially so that we can accommodate presence in potentially remote areas, LEOScope supports not only traditional university-based nodes as PlanetLab did, but also nodes in volunteers' home networks.

The key consideration here is that some networking experiments such as iperf can make heavy usage of the network connection, which should not interfere with the *primary users*, i.e., the residential volunteers. To this end, we develop a **scavenger mode** inspired by scavenger transport protocols [41, 51]. When enabled for a node, this treats experiments as *secondary users* and stops the experiments running when the primary user starts using the network.

This ensures, for example, that when the volunteers are using their home network for an important work-related teleconference, or even for watching Netflix, their connection quality is not impaired by a 'heavy' experiment.

To support the above, we use the telemetry features available by default on Starlink, which gives us an estimate of the current *total* traffic t_{all} being used by the entire node, both by the experiments as well as the volunteers' own network traffic. Using the Docker API, the node can also determine the total traffic t_{expt} being used by all the running experiment containers on the node, which allows it to calculate the amount of the volunteers' own (non-experiment) traffic as $t_{all} - t_{expt}$. If this exceeds a configurable threshold Δ , then the node infers that the volunteer is using their network, and stops the running experiments one by one in descending order of current network usage, until $t_{all} - t_{expt}$ falls below Δ . Note that this design allows multiple 'lightweight' experiments (e.g., ping) to continue running, so long as they do not impinge on the volunteers' network usage, i.e., $t_{all} - t_{expt} < \Delta$. An observer periodically checks and reschedules the stopped experiments after the volunteers' flows finish. We also allow volunteers to set monthly limits on how much data they wish to allow LEOScope experiments to consume as a whole, across all experiments using their node. Volunteers can use this feature if they wish to ensure that LEOScope does not place unfair demands on their LEO network provider.

Many have observed that LEO connection quality can vary dynamically [31, 42, 50]. Recognizing that researchers might want to schedule experiments based on *exogenous factors*, such as rain or snow which can affect connectivity, or directly based on *endogenous factors* such as periods of high latency or low throughput, we build

the **trigger mode**. The trigger mode operates similar to the “If This Then That (IFTTT)” paradigm [4]. When a certain condition (“if this”) such as heavy rain or high latency occurs, a “then that” experiment (e.g., measure throughput) can be specified to execute.

Whereas the trigger mode is useful if it is not known a priori when to schedule an experiment, researchers may also want to schedule experiments at pre-specified time points: For instance, at regular time periods throughout the day; or during evening usage peaks. This is implemented by a **cron mode**, which allows experiments to be scheduled at pre-specified times, using crontab syntax. Cron mode also helps schedule experiment repeats to ensure statistical validity.

Although experiments are isolated from each other in docker containers, they all use the same outgoing Starlink connection and some experiments may still interfere with each other: For example, to explore bandwidth available during satellite handoffs, an experiment may probe to see achievable throughput, and its results will be affected by other experiments which may be running at the same time. To accommodate such use cases, we allow experiments to mark themselves as **bandwidth sensitive**. We make sure to schedule such experiments in a mutually exclusive fashion.

4 ARCHITECTURE DETAILS

This section describes the LEOScope building blocks (Fig.1), and how these support the design considerations of §3.

Researchers are the users of the testbed. Using a web UI, they can submit network and application-level experiments (Procedure described in Appendix (A)), request to schedule them in various ways (at a single specific time point, repeatedly, using cron syntax, or based on a user-defined trigger), select a set of measurement clients and servers to run these experiments on, and view the results dashboard.

Cloud Orchestrator ensures that researchers can conduct experiments and gather data on the testbed without directly interacting with the deployed nodes. As depicted in Fig.1(b), the orchestrator is responsible for scheduling experiments across measurement clients and servers, managing schedule conflicts, and monitoring node health. It operates two core services at scale: scheduling experiments and managing measurement servers, client nodes, and user access.

The orchestrator maintains a set of data stores (bottom right in Fig. 1(b)) that record details about nodes, their health status, and user information. It also tracks scheduled experiments and node mappings, while consolidating data received from individual clients and servers into the results store. When a new experiment scheduling request is received, the orchestrator evaluates it against existing schedules to decide whether to accept or reject it. Additionally, individual measurement clients can autonomously reschedule experiments to facilitate scavenger mode. In such cases, the measurement clients notify the orchestrator of such changes.

The key asset provided by the testbed is access to a geographically distributed set of **Measurement Clients**. A client node consists of an LEO user terminal, and a host device, which could range from standard workstations and Raspberry Pi [9] to more advanced edge devices like Azure Stack Edge [43] or Intel NUC [29]. These host devices sit behind the LEO terminal and can access

the Internet via that terminal. The client nodes register with the cloud orchestrator and maintain communication through regular heartbeat signals to monitor node health.

When an experiment schedule involves a particular client node, the orchestrator first checks the node’s health and reviews the existing schedule. If free slots are available, it sends a new experiment request to the client. The client then prepares by downloading the necessary artifacts and setting up the local environment. Because all measurement clients are located behind NATs and firewalls, the cloud orchestrator piggybacks experiment schedules and other commands onto these heartbeat responses. This ensures seamless communication with both clients and servers.

Logistically, the extreme flexibility offered by LEOScope means that we cannot predict all the requirements of specific individual experiments. Thus, experiments may need to provision additional components for themselves, such as the two components highlighted with red dashed boxes in Fig. 1(a): First, **Measurement Servers** may need to be deployed on the public cloud which depends on the clients’ node setup to facilitate client-server measurement experiments such as iperf [6]. Second, nearly all experiments need some storage for their results. Since multiple experiments can be running on each node, and each node can generate arbitrary amounts of data, we do not provide permanent storage, and experiments need to push their data to **Cloud Storage**.

LEOScope provides special support for both measurement servers and (Azure) cloud storage. Experiments can specify measurement servers, which then receive schedules from the cloud orchestrator, download necessary artifacts, and set up the local environment before client requests arrive. Once experiments are conducted, the servers push the results back to the cloud storage for passive analysis. Experiments running on the measurement clients can also push data to cloud storage. The storage is hierarchically structured into directories such that it is straightforward to navigate to a specific instance of an experiment on a measurement node. The cloud orchestrator leverages a post-processing library that fetches data from this cloud storage and generates insights (plots, visuals, etc.) to be displayed on the user dashboard.

Measurement clients/servers run four core services as shown in Fig.1(b). (1) A scheduler service that processes experiment schedules from the cloud orchestrator and updates the local cron daemon accordingly. (2) An experiment executor container that conducts network experiments based on schedules approved by the orchestrator. (3) A terminal data consumer service that continuously collects disclosed data from the user terminal, which aids in further analysis and potentially triggers additional scheduling. (4) A heartbeat service to inform the orchestrator that it is alive.

5 EVALUATION

Having described the architecture in detail, we next evaluate whether the implementation matches the needs raised by the design considerations (§3). In each case, Fig. 2 shows one specific node/location as proof-of-concept, but each effect has been verified for multiple locations.

To test **Scavenger mode**, we instrument the Edinburgh node (EDN) by scheduling a high bandwidth experiment (bottom panel (blue) in Fig. 2(a), shows throughput) and interrupt this with the

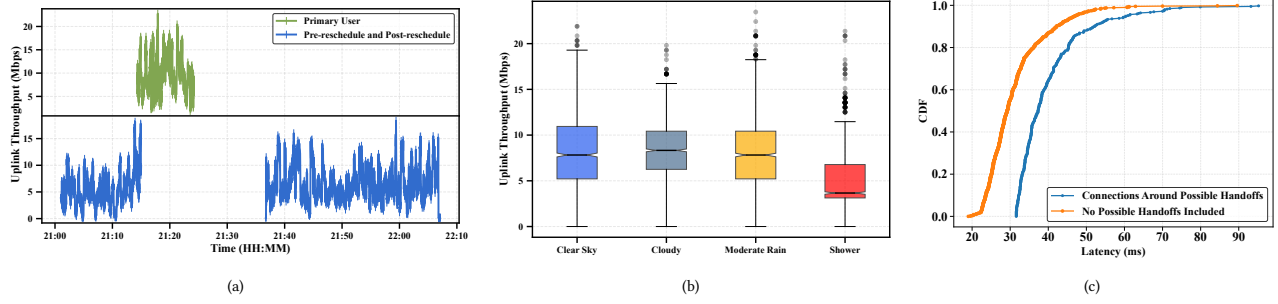


Figure 2: (a) Scavenger Mode timeline when primary user starts to use bandwidth while experiments are running (EDN) (b) Triggering throughput measurements for different weather conditions (VIC) (c) Scheduling cron jobs RTT measurement for times when satellites go out of sight. (LON)

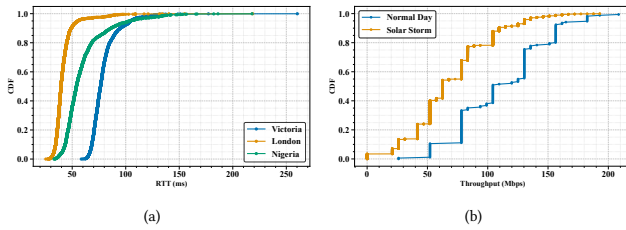


Figure 3: Need for geographic diversity: (a) RTT CDF in London, UK Victoria, CA and Abuja, NG show different latency profiles. (b) Throughput in Edinburgh under influence of Solar Storm on May 10 2024 showcases benefits of distributed global presence.

primary user starting to use bandwidth (top panel, green). Notice that the experiment is stopped immediately and rescheduled. The experiment resumes again after the primary user has stopped.

Fig. 2(b) demonstrates the utility of **trigger mode**, by using the OpenWeather API [7] to schedule throughput measurements for different weather conditions in Victoria, British Columbia. Similar to Kassem *et al.* [31, Fig. 4] who demonstrated that latency (specifically, the time taken to fetch the contents of a web page) increases during rainy conditions, this shows that throughput also decreases during heavy showers, compared to clear sky or even moderate rain.

While the result of Fig. 2(b) is somewhat expected given the results of [31], it demonstrates how experiments can be *proactively* programmed to collect data based on exogenous factors such as weather. In contrast, [31] first collected data relating to “page transit time” (a measure related to overall connection latency) from a browser extension, and later on, used correlations with the weather as on the date of data collection to demonstrate the effect of weather. Our proactive approach allows us to directly gather throughput or latency (not shown) data, and more importantly, guarantees we can gather sufficient data for statistical validity, which might not be feasible for a given location or time of interest, with the post hoc approach of [31].

To showcase **cron-like scheduling** of experiments when the times of interest can be calculated a priori, we ask whether users’ throughput is affected when satellites go out of line-of-sight at a

given location. Given a Starlink terminal’s location, we can use the Celestrak API [3] to compute the list of satellites overhead at any given time point and calculate the time slots when these satellites go out of sight for the terminal, using Starlink’s FCC filings [55], which indicate that satellites are considered out-of-sight when the elevation angle is lower than 25 degrees.

Fig. 2(c) shows the results, focusing on London, UK. Whilst we cannot say whether the user terminal is in fact connected to a satellite which goes out of line-of-sight, our hypothesis is that at *some* of those time points, the terminal may in fact be connected, and the concomitant handover will affect throughput. To test this, Fig. 2(c) compares the Cumulative Distribution of the throughputs achieved at all time points when a satellite goes out of sight with the throughput achieved at other time points. As expected, we find that the throughput is in fact decreased at potential handover points. Although in line with what is expected, we believe this is a new result: it has, to our knowledge, not been experimentally confirmed before in other measurement works.

Fig 3 showcases the need for **geographic diversity** in a LEO Testbed. We first show (Fig. 3(a)) that the distribution of latency (RTT) can vary across different geographic locations, suggesting that any proposals for adapting network protocols to the dynamically changing latencies of LEO megaconstellations should be tested in multiple locations. Fig. 3(b) shows how the May 2024 G5 geomagnetic storm (the highest severity solar storm possible) in various locations around the world, demonstrated the effect of such storms on Starlink. Fig. 3(b) shows that the throughput decreased significantly during the storm at one location (Edinburgh). Similar results were obtained as far south as Victoria, British Columbia and London, UK, which also saw aurora displays.

6 SUMMARY AND FUTURE OUTLOOK

LEOScope has been designed to be adapted to the unique needs of LEO connectivity measurements – First and foremost, such a testbed needs to enable a global footprint, including remote areas. For this, it is essential for the testbed to be able to work with nodes that are installed in volunteers’ homes, where the LEO connection might be their sole source of connectivity. To accommodate this, we designed the *scavenger mode*, which pauses researchers’ experiments when the primary users (the volunteers) start to use the network. Second, unlike the more stable terrestrial broadband

connections, modern LEO networks are much more dynamic in connection quality, and it is important that experiments have an efficient way to run experiments when events of interest occur. For this, we designed the *trigger mode*, which operates similar to the “If This Then That” paradigm, triggering an experiment when an endogenous (e.g., increased latency or packet drops) or exogenous (such as rain or snow) event of interest occurs. LEOScope also has support for easily adding experiments using a Web UI, and storing results on cloud storage, which can be retrieved for analyses later on. We also have support for other functionality such as cron-style scheduling of experiment repeats for statistical validity or temporal coverage, and mutually exclusive scheduling of *bandwidth sensitive* experiments.

While LEOScope allows researchers to define almost any experiment they want, many may simply want “basic” data such as throughput and latency at each node. We provide this through a dashboard at <https://leoscope.surrey.ac.uk/dashboard>, which also gives live node statuses.

We believe, based on our own experience and those of researchers who are early users of LEOScope, that the currently implemented set of features already make it a valuable platform for LEO network measurement research. However, we are also planning on adding the following additional features, and are open to community suggestions on features they might like to see:

- Although currently LEOScope footprint comprises solely of Starlink terminals, the essential architecture (Fig. 1) is agnostic to the actual network, and can accommodate multiple constellations. We are highly interested in expanding to other constellations, including GEO and MEO networks which have different performance [50]. GEO and LEO vary in terms of the coverage offered. It is important to be able to study and understand such phenomena and trade-offs as LEO connectivity grows. Some nodes might be multi-homed and offer connectivity to multiple (kinds of) networks. We will need to develop new network fingerprinting methods to help researchers identify the networks being used.
- We are equally interested in growing our geographic footprint. Although we currently have presence three continents (a fourth will be added soon), coverage is still sparse. For instance, we only have one node in all of Africa, in Abuja, Nigeria. This node also suffers from frequent power outages which makes scheduling experiments difficult. We encourage contributions of nodes globally, to grow our global footprint. In the near future, we envision providing regular reports of useful information such as this across all nodes, similar to RIPE Atlas data provided across its probes [10].
- Researchers currently also have to design their own measurement servers for client-server experiments, although we provide base code for some common experiments such as *iperf*. A closer integration with server-based approaches such as M-Lab [23] can be a convenient way to merge the client-side approaches such as ours with server-side measurement tools such as M-Lab’s NDT. A simple enhancement could include a docker *measurement client* deployed on LEOScope nodes, which uses M-Lab servers as the corresponding *measurement server* (see Fig. 1(a)).

ACKNOWLEDGMENTS

LEOScope has benefited from the contributions of many dedicated individuals and institutions. We are especially grateful to the volunteers who hosted nodes and to the researchers who actively used LEOScope in their work and offered insightful early feedback that guided its initial development.

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APPENDICES

We are building LEOScope for democratising LEO Satellite Network (LSN) measurements and driving LEO network innovation at scale. In the main paper, we demonstrated LEOScope's unique features necessary for capturing LEO network dynamics and onboarding of volunteers and also reported initial measurements to showcase some of its capabilities. Given the short window to influence this new kind of network before they already start serving a significant fraction of the Internet traffic, this paper is also a call to the larger community to join this effort and scale its footprint. We invite researchers to join our testbed by signing up at <https://leoscope.surrey.ac.uk> and contributing additional nodes if possible. We also invite a discussion with the community on how best to manage and run LEOScope as a community resource, by emailing us on leoscopetestbed@surrey.ac.uk or joining a Slack channel for researchers interested in LEO Networking Research².

We include a set of appendices below that provide details of our processes for how experimenters can use LEOScope (Appendix A), where the nodes are hosted currently and how to contribute a new node (Appendix B), and a discussion of how to promote ethical and responsible experimentation that does not compromise or cause harm to the volunteers contributing new nodes (Appendix C). We view these as evolving processes and welcome a discussion with the community on evolving best practices that work for all. Appendix D provides a comparative analysis of LEOScope against existing platforms such as RIPE Atlas, M-Lab, and PlanetLab, highlighting LEOScope's LEO-specific capabilities such as satellite telemetry integration and event-based measurement triggering. We view these appendices – and LEOScope itself – as evolving efforts, and we invite the community to engage with us in refining best practices, improving infrastructure, and collectively advancing LEO network research.

²<https://leoscope.surrey.ac.uk/researchers-channel>

A HOW TO RUN EXPERIMENTS

Experimenters wishing to use LEOScope need to create an account on the LEOScope website (<https://leoscope.surrey.ac.uk>). We have created a document³ for the steps that experimenters can follow to set up an experiment wrapped up as a docker container. Crucially, experiment execution hinges on a YAML configuration file⁴ detailing cloud result destinations and experiment runner specifics. We also provide an example experiment⁵ that runs Ping tests and can be used as a template to develop other experiments. The ping experiment runner, available on DockerHub, is currently deployed as a Docker image on all the nodes. We expect that, over time, a library of commonly used experiments will be made available for reuse by the community as a catalog of docker images.

B HOW TO CONTRIBUTE NODES

The testbed is currently at a nascent stage. As depicted in the map in Fig. 4, LEOScope currently comprises nodes across various locations which are shown in red color, enabling researchers to conduct experiments on a geographically diverse infrastructure.

We are actively looking to enhance this footprint and will be onboarding three new nodes in the coming months. We would warmly welcome any additional nodes from the SIGCOMM community, including from countries already having a LEOScope presence. Contributing a node is relatively simple, and involves provisioning a small computer such as an Intel NUC behind a Starlink terminal, registering the node on the LEOScope website, and cloning and running the testbed repository⁶. We provide a document that details all the steps⁷.

Note that volunteers are completely free to withdraw their nodes from the testbed whenever it proves inconvenient to them. Indeed, nodes can also temporarily withdraw, for example, if they wish to power down their router and pause Starlink connectivity when going on vacation. The User Interface also allows volunteers to attach their nodes to LEOScope during specific times if they wish (e.g., from 1AM-4AM local time). To further optimize resource management and mitigate costs associated with Starlink usage on volunteered nodes, we have also implemented a billing feature that tracks and monitors measurement usage. This feature records usage metrics in a key-value (KV) structure, enabling real-time tracking of resource consumption by researchers conducting experiments on the testbed. During the initial setup of a node for integration into LEOScope, the node administrator can define monthly usage limits tailored to the node's capacity and operational constraints. These limits are enforced to ensure fair and sustainable resource allocation across all users. At the end of each month, the usage counter is reset, allowing researchers to continue their measurements within the predefined thresholds. This approach not only promotes cost efficiency but also ensures transparency and accountability in resource utilization across the testbed.

³<https://leoscope.surrey.ac.uk/experiment-running-guideline>

⁴https://github.com/leoscope-testbed/experiment_config/blob/main/experiment-config.yml

⁵https://github.com/leoscope-testbed/ping_experiment_guideline

⁶<https://github.com/leoscope-testbed/global-testbed>

⁷<https://leoscope.surrey.ac.uk/node-setup-guideline>



Figure 4: Existing Nodes on the Testbed

C ENSURING RESPONSIBLE USAGE

Because we allow almost any experiment to be run, we believe ethics and responsible research questions need to be carefully considered, especially in nodes located in the homes of layperson volunteers. To ensure the volunteers are not disadvantaged in using their own network, we implemented the scavenger mode (§3) to enforce that researchers' experiments do not interfere the volunteers primary usage.

To mitigate risks from unauthorized Docker containers on a Starlink-connected Linux machine, we have implemented strict network segmentation and least-privilege principles. Our Node Application setup is isolated on separate VLANs or subnets on the Linux machine connected to the Starlink Dish. Docker has been hardened by disabling `--network=host` and configuring custom bridge networks to prevent access to the local LAN. Containers are run with minimal privileges, dropping all unnecessary capabilities (`--cap-drop=ALL`) and using read-only filesystems to limit potential attack vectors. Access to the Docker socket (`/var/run/docker.sock`) is restricted to root-only, and all container deployments are audited to ensure no unnecessary ports are exposed or sensitive host directories are mounted.

Going well beyond this, we also require researchers to sign a EULA, laying down legally enforceable responsibilities to not cause harm to the testbed resources, including the volunteer nodes. Our EULA also provides legal mechanisms to protect against misuse and has been designed by the legal team at University of Surrey as well as vetted by the legal teams of other co-authors' partner institutions.

To further remind researchers of their responsibilities, we have an onboarding procedure and document for all researchers, which asks them to be mindful of and respectful towards the needs of the volunteer and university node hosts, including their privacy. For example, researchers are expressly forbidden from accessing the private home networks of the volunteers, or deanonymising the nodes through either their IP address, or the location information.

There is still a risk that some researchers might ignore or forget our instructions and compromise volunteers' privacy in one of the many experiments they may run. Thus, for each new experiment, we ask for a PDF of an Institutional Review Board (IRB) approval to be uploaded, or, as a more lightweight option, experimenters can answer a series of questions that if answered in the negative, gives reasonable assurance that the experiment will not compromise or

harm the nodes they are running on. The hope is that by going through a checklist of questions, or by virtue of going through the process of satisfying an IRB, experimenters can think more consciously about the potential for harm and through that also avoid making inadvertent mistakes that could compromise the volunteer (or university-hosted) nodes. We make a copy of the list of questions available without logging in to LEOScope here⁸.

The above text only considers responsible usage by *researchers* conducting experiments on the testbed. Note that the current design of LEOScope implicitly trusts that the volunteers hosting a testbed node. We believe this is an acceptable solution for now because we have personal relationships with all the hosts. However, as LEOScope scales, we will need to implement mechanisms to provide assurance to researchers that they are indeed on a Starlink connection for example (e.g., some nodes might have fall back connections or move away from Starlink subscriptions without notifying LEOScope. To mitigate such risks, we would need to implement checks on the orchestrator side, e.g., to ensure that the node is behind an IP address assigned to Starlink.

D COMPARATIVE ANALYSIS OF LEOSCOPE WITH EXISTING TOOLS & TESTBEDS

LEOScope offers a unique combination of features specifically designed for LEO satellite network research. While established platforms like RIPE Atlas [10] and M-Lab [23] provide valuable general-purpose measurement capabilities, they lack the specialized features required for comprehensive LEO satellite network studies. Planet-Lab [8] and EdgeNet [20] offer greater experiment flexibility but lack the satellite-specific telemetry and triggering mechanisms essential for correlating network performance with orbital dynamics. Monroe [12] provides excellent support for cellular network research with built-in telemetry but, like other platforms, was not designed with the unique characteristics of LEO satellite networks in mind. LEOScope builds upon lessons learned from these platforms while introducing novel capabilities specifically tailored to the challenges of LEO satellite network research:

- (1) **Integration of network and satellite telemetry:** LEOScope uniquely combines network performance measurements with satellite positioning data, enabling researchers to correlate performance variations with specific orbital conditions.
- (2) **Event-based triggering:** The platform can initiate measurements based on satellite events (e.g., handovers, specific elevation angles), allowing researchers to target phenomena of interest with precision.
- (3) **Resource-aware operation:** LEOScope's comprehensive scavenger mode ensures that experiments do not interfere with volunteers' primary network usage, encouraging broader participation without compromising measurement quality.
- (4) **Hardware flexibility:** By supporting diverse hardware configurations, LEOScope lowers the barrier to entry for volunteers while maintaining a consistent measurement methodology.

This combination of features positions LEOScope as the first measurement platform specifically designed for the unique challenges of LEO satellite networks, addressing a critical gap in existing research infrastructure.

⁸<https://leoscope.surrey.ac.uk/ethical-form>